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On Measuring Growth and Inequality

Components of Changes in Poverty

with Application to Thailand

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ON MEASURING GROWTH AND INEQUALITY
COMPONENTS OF POVERTY
WITH APPLICATION TO THAILAND

by

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Abstract

This paper is concerned with the decomposition of change in poverty into growth and inequality components. A new decomposition procedure is derived using an axiomatic approach. This methodology is applied to explain changes in poverty in Thailand during the years 1988, 1990, 1992 and 1994.

1. Introduction

It is of considerable interest to know whether overall economic growth reduces poverty, i.e., is there a trickle-down mechanism. Has economic growth been accompanied by increased inequality of income. To understand the impact of economic growth on poverty, we should measure separately the impact on poverty, over time, of changes in average income and in its inequality or in other words we must decompose the total change in poverty into: (1) the impact of growth when the distribution of income does not change and (2) the effect of income redistribution when the total income of the society remains unchanged.

The decomposition of poverty into growth and inequality effects has recently attracted much attention (Kakwani and Subbarao (1990, 1991, 1992), Datt and Ravallion 1992 and Jain and Tendulkar (1990)). The main objective of the present paper is to derive a new decomposition procedure using an axiomatic approach. These axioms provide a basis for evaluating the alternative decomposition procedures proposed earlier. It is demonstrated that the earlier procedures violate some intuitively natural axioms. We introduce the idea of average growth and inequality effects, the sum of which is equal to the total change in poverty. Thus, we propose an exact decomposition in which growth and inequality effects are evaluated consistently. The methodology of decomposition developed on the basis of bilateral comparisons is then extended to the situations of multilateral comparisons which allows us to measure the growth and inequality effects in a symmetric manner when there are more than two time periods.

The methodology developed in the paper is applied to analyze changes in poverty in Thailand covering the periods from 1988 to 1994. The four socioeconomic surveys conducted in Thailand during the years 1988, 1990, 1992 and 1994 are utilized for this purpose.

2. Decomposition Axioms

Suppose θ is a poverty measure which is fully characterized by the poverty line income, the mean income and the Lorenz curve (which is a general measure of relative income inequality), then

$$\theta = \theta(z, \mu, L(p))$$

where z is the poverty line, μ is the mean income of the society and $L(p)$ is the equation of the Lorenz curve; p varies from 0 to 1.

Since households differ in size, age composition, and other characteristics, it is expected that they will have different needs. We assume that the household income which is the basis for computing μ and $L(p)$ has been adjusted for the differing needs of the households.

We wish to explain the change in poverty between period i (base year) to period j (terminal year) in terms of growth and inequality components. Let us represent the change in poverty between i and j as θ_{ij} :

$$\theta_{ij} = \theta(z, \mu_j, L_j(p)) - \theta(z, \mu_i, L_i(p)) \quad (1)$$

where μ_i and μ_j are the mean incomes in the years i and j , respectively and $L_i(p)$ and $L_j(p)$ are the Lorenz curves for the years i and j , respectively. The mean incomes μ_i and μ_j have been adjusted for price changes between the two periods. We call θ_{ij} as the total poverty effect.

The pure growth effect is defined as the change in poverty if the mean income were to change but the relative income distribution measured by the Lorenz curve remained unchanged. Similarly, the pure inequality effect is defined as the change in poverty if the Lorenz curve were to change but the mean income at the constant prices remained the same. Let us denote the growth effect (between years i and j) by G_{ij} and the inequality effect by I_{ij} , then we postulate that there must exist a relationship between the total poverty effect on the one hand and the growth and inequality effects on the other hand. Thus, we have

$$\theta_{ij} = f(G_{ij}, I_{ij}) \quad (2)$$

where the function form is determined by the following axioms.

Axiom 1: If $I_{ij} = 0$, then $\theta_{ij} = G_{ij}$ and if $G_{ij} = 0$, then $\theta_{ij} = I_{ij}$.

This axiom is intuitively natural. It implies that when the growth effect (inequality effect) is zero, then the change in poverty must be entirely due to change in income inequality (mean income). This axiom immediately leads to the following weaker axiom.

Axiom 1A: If both the G_{ij} and I_{ij} are equal to zero, then θ_{ij} must also be equal to zero.

Axiom 2: If $G_{ij} \leq 0$ and $I_{ij} \leq 0$, then $\theta_{ij} \leq 0$ and if $G_{ij} \geq 0$ and $I_{ij} \geq 0$, then $\theta_{ij} \geq 0$.

This axiom does not need much of an explanation. It implies that if both growth and inequality effects are less (greater) than or equal to zero, then the total poverty effect must also be less (greater) than zero. If, however, $G_{ij} \leq 0$ (≥ 0) and $I_{ij} \geq 0$ (≤ 0), then the total change in poverty must depend on the magnitude of G_{ij} and I_{ij} .

Note that $G_{ij} \geq 0$ (≤ 0) implies that the change in mean has an adverse (favourable) effect on the poor which can happen only if the mean (real) income decreases (increases) between the periods i and j . Similarly $I_{ij} \geq 0$ (≤ 0) implies that the change in inequality has an adverse (favourable) effect on the poor. Following this terminology, it can be seen that Axiom 2 will always be satisfied if

$$\frac{\partial f}{\partial G_{ij}} \geq 0 \text{ and } \frac{\partial f}{\partial I_{ij}} \geq 0 .$$

From (1), we note that

$$\theta_{ij} = -\theta_{ji}$$

will always hold. This leads us to propose the following axiom

Axiom 3: $G_{ij} = -G_{ji}$ and $I_{ij} = -I_{ji}$.

G_{ij} and I_{ij} are the growth and inequality effects when we go from base year i (characterized by μ_i and $L_i(p)$) to the terminal year j (characterized by μ_j and $L_j(p)$) and, therefore, they should be of the same magnitude and of opposite sign when going from terminal year j to base year i . It means that if poverty has increased by say five points because of pure growth effect, when we go from terminal year j to base year i must imply a reduction in poverty by exactly five points. The same argument must apply to the inequality effect. Thus, both growth and inequality effects must be symmetric with respect to base and terminal years. The violation of Axiom 3 will give rise to the problem of nominating the base and terminal years as the reference. And that can only be done on an ad hoc basis.

3. An Evaluation of Alternative Approaches

In view of the axioms proposed in the previous section, we evaluate the alternative decomposition procedures proposed in the literature.

Kakwani and Subbarao (1990) proposed the following decomposition procedure.

$$G_{ij} = \theta(z, \mu_j, L_i(p)) - \theta(z, \mu_i, L_i(p)) \quad (3)$$

which is the change in poverty if the mean changed from μ_i to μ_j but the Lorenz curve in the base year i remained fixed. They derived the inequality effect as the residual:

$$I_{ij} = \theta_{ij} - G_{ij} \quad (4)$$

It can be seen that Axioms 1, 1A, 2 are always satisfied by this decomposition procedure. To see if Axiom 3 is satisfied, write

$$G_{ji} = \theta(z, \mu_i, L_j(p)) - \theta(z, \mu_j, L_j(p)) \quad (5)$$

which shows that $G_{ij} = -G_{ji}$ will hold only if $L_i(p) = L_j(p)$. Thus, Axiom 3 is violated.

The pure inequality effect must be defined as the change in poverty if the Lorenz curve were to change but the mean (real) income remained unchanged. The inequality effect in (4) does not possess such an interpretation. In a subsequent paper, Kakwani and Subbarao (1990a) computed the inequality effect as

$$I_{ij} = \theta(z, \mu_i, L_j(p)) - \theta(z, \mu_i, L_i(p)) \quad (6)$$

If we define G_{ij} as in (3) and I_{ij} as in (6), then there will not exist an exact relationship between θ_{ij} , G_{ij} and I_{ij} . This prompted Datt and Ravallion (1992) to propose the following decomposition

$$\theta_{ij} = G_{ij} + I_{ij} + R_{ij} \quad (7)$$

where R_{ij} is the residual term which is given by

$$R_{ij} = -G_{ji} - G_{ij} \quad (8)$$

Datt and Ravallion (1992) have found that the residual term in (8) can be quite large (even larger than the inequality effect). Since there are only two factors (mean income and inequality) which explain the total change in poverty, what meaning can be attached to such a high value of residual is not clear.

Jain and Tendulkar (1991) have proposed two alternative decompositions. Define

$$G_{ij}^* = \theta(z, \mu_j, L_j(p)) - \theta(z, \mu_i, L_j(p))$$

and

$$I_{ij}^* = \theta(z, \mu_j, L_j(p)) - \theta(z, \mu_j, L_i(p))$$

then their decompositions are written as

$$\theta_{ij} = G_{ij} + I_{ij}^* \quad (9)$$

and

$$\theta_{ij} = G_{ij}^* + I_{ij} \quad (10)$$

Although Jain-Fendulkar's decompositions are exact with no residual term but in our view it makes little sense in defining the pure growth effect in terms of the base (terminal) year Lorenz curve and the pure inequality effect in terms of the terminal (base) year mean income. One should fix either the base income distribution (characterized by the base year mean income and the base year Lorenz curve) or the terminal year income distribution (characterized by the terminal mean income and terminal Lorenz curve). But they choose the base year mean income for the pure inequality effect and terminal Lorenz curve for the pure growth effect and vice versa. In our view, it is important to use the same reference periods for defining the pure growth and inequality effects. Despite these drawbacks, Jain-Tendulkar decompositions satisfy axioms 1, 1A and 2. But axiom 3 is violated by their decompositions. Furthermore, which one of two decompositions should be chosen is a difficult question that has not been addressed.

4. A New Decomposition Derived

$\theta_{ij}(\theta_{ji})$ is the change in poverty when we go from year i (year j) to year j (year i), obviously then

$$\theta_{ij} = -\theta_{ji}$$

which in view of (2) implies

$$f(G_{ij}, I_{ij}) = -f(G_{ji}, I_{ji})$$

which on using Axiom 3 becomes

$$f(G_{ij}, I_{ij}) = -f(-G_{ij}, -I_{ij}) .$$

This equation will always hold if f is a homogeneous function of degree one in G_{ij} , I_{ij} . Then using Euler's theorem, we obtain

$$f(G_{ij}, I_{ij}) = \frac{\partial f}{\partial G_{ij}} G_{ij} + \frac{\partial f}{\partial I_{ij}} I_{ij} \quad (11)$$

Then using Axiom 1, if $I_{ij} = 0$, $f(G_{ij}, I_{ij} = G_{ij})$ which, on substituting in (11), gives

$$G_{ij} = \frac{\partial f}{\partial G_{ij}} G_{ij} \quad (12)$$

and similarly, if $G_{ij} = 0$, $f(G_{ij}, I_{ij}) = I_{ij}$, then

$$I_{ij} = \frac{\partial f}{\partial I_{ij}} I_{ij} \quad (13)$$

Equations (12) and (13) will hold only if

$$\frac{\partial f}{\partial G_{ij}} = \frac{\partial f}{\partial I_{ij}} = 1$$

Thus, $f(G_{ij}, I_{ij})$ must be of the form

$$\theta_{ij} = f(G_{ij}, I_{ij}) = G_{ij} + I_{ij} \quad (14)$$

which shows that the total poverty effect is equal to the sum of growth and inequality effects.

This equation allows to analyze how much of the total change in poverty is explained individually by the growth and inequality effects.

Equation (14) will always satisfy axioms 1, 1A and 2. Now we need to define G_{ij} and I_{ij} so that axiom 3 and equation (14) are always satisfied.

Recall that G_{ij} in (3) (I_{ij} in (6)) is the change in poverty due to a change in mean income (Lorenz curve) while holding the base year i Lorenz curve (mean income) constant. One can similarly define the growth and inequality effects by keeping terminal year j Lorenz curve and terminal year j mean income constant, respectively. The question is whether we use the base or terminal year distributions to define the growth and inequality effects. Datt and Ravallion (1992) regard the base year distribution to be a natural choice of a reference. We take the view that the base year distribution is not better or worse than the terminal year

distribution as a reference. There exists no reason to introduce an asymmetry in the treatment of base and terminal years. Thus, the growth and inequality effects should be symmetric with respect to the base and terminal years: our axiom 3 which is indeed intuitively natural implies a symmetric treatment of the base and terminal years. This leads us to define the average growth and inequality effects as given by

$$\hat{G}_{ij} = \frac{1}{2} \left[\theta(z, \mu_j, L_i(p)) - \theta(z, \mu_i, L_i(p)) + \theta(z, \mu_j, L_j(p)) - \theta(z, \mu_i, L_j(p)) \right] \quad (15)$$

and

$$\hat{I}_{ij} = \frac{1}{2} \left[\theta(z, \mu_i, L_j(p)) - \theta(z, \mu_i, L_i(p)) + \theta(z, \mu_j, L_j(p)) - \theta(z, \mu_j, L_i(p)) \right] \quad (16)$$

It can be observed that

$$\hat{G}_{ij} = -\hat{G}_{ji}$$

and

$$\hat{I}_{ij} = -\hat{I}_{ji}$$

which imply that axiom 3 is always satisfied.

Combining (15) and (16) gives

$$\theta_{ij} = \hat{G}_{ij} + \hat{I}_{ij} \quad (17)$$

which shows that the total change in poverty is equal to the sum of average growth and inequality effects which implies that axioms 1, 1A and 2 are satisfied. Thus, our proposed growth and inequality effects (given in (15) and (16), respectively) will always satisfy all the axioms proposed above.

5. When There Are More Than Two Periods

So far our methodology of measuring growth and inequality effects has been based on the assumption that there are only two periods of comparison, i.e. between periods i and j .

These are the bilateral comparisons. Generally, we wish to make multilateral comparisons when there are more than two periods. In this section, we propose a generalization of the analysis presented so far.

Suppose there are n periods for which we want to explain changes in poverty. It is easy to see that θ_{ij} as defined in (1) will always satisfy

$$\theta_{ij} = \theta_{ik} + \theta_{kj}$$

for all values of $k = 1, 2, \dots, n$. This prompts us to propose the following two axioms.

Axiom 4: $G_{ij} = G_{ik} + G_{kj}$ for all $k = 1, 2, \dots, n$.

Axiom 5: $I_{ij} = I_{ik} + I_{kj}$ for all $k = 1, 2, \dots, n$.

Axioms 4 and 5 imply that the growth and inequality effects should be transitive. Since the total poverty effect is always transitive, it is reasonable to require that the growth and inequality effects (which are the individual components of total change in poverty) should also be transitive. These are the most fundamental axioms in the situations of multilateral comparisons. These transitivity requirements imply that the growth and inequality effects for the sub-periods add up to those for the period as a whole. This property will not hold for the growth and inequality effects measured on the basis of bilateral comparisons. In order to satisfy these transitivity axioms, Datt and Ravallion (1992) suggested to use a fix reference year for all decomposition periods. This solution is not satisfactory because the results will be sensitive to the choice of reference year. Although they suggest that the initial date of the first decomposition period is a natural choice, this is also arbitrary.

We would like to have measures of growth and inequality effects which satisfy all the axioms (1, 1A, 2, 3, 4 and 5) presented in the paper. This can be accomplished by making all possible pairwise comparisons. Thus, we propose the new measures of growth and inequality given by $\tilde{G}_{ij}$ and $\tilde{I}_{ij}$, respectively:

$$\tilde{G}_{ij} = \frac{1}{n} \sum_{k=1}^n (\hat{G}_{ik} + \hat{G}_{kj}) \quad (18)$$

and

$$\tilde{I}_{ij} = \frac{1}{n} \sum_{k=1}^n (\hat{I}_{ik} + \hat{I}_{kj}) \quad (19)$$

where i, j vary from 1 to n ; $\hat{G}_{ij}$ and $\hat{I}_{ij}$ are as defined in (15) and (16), respectively.

We may now evaluate whether or not $\tilde{G}_{ij}$ and $\tilde{I}_{ij}$ will satisfy all the axioms proposed in the paper.

First, let us write from (18)

$$\tilde{G}_{ji} = \frac{1}{n} \sum_{k=1}^n (\hat{G}_{jk} + \hat{G}_{ki})$$

which on utilizing $\hat{G}_{ij} = -\hat{G}_{ji}$ immediately gives $\tilde{G}_{ij} = -\tilde{G}_{ji}$. Similarly, it can be proved that

$\tilde{I}_{ij} = -\tilde{I}_{ji}$. Thus, our proposed measures $\tilde{G}_{ij}$ and $\tilde{I}_{ij}$ will always satisfy axiom 3.

Adding $\tilde{G}_{ij}$ and $\tilde{I}_{ij}$ and utilizing (17) gives

$$\theta_{ij} = \tilde{G}_{ij} + \tilde{I}_{ij}$$

which demonstrates that $\tilde{G}_{ij}$ and $\tilde{I}_{ij}$ will satisfy axioms 1, 1A and 2.

Finally, utilizing (18), write

$$\tilde{G}_{i\ell} = \frac{1}{n} \sum_{k=1}^n (\hat{G}_{ik} + \hat{G}_{k\ell})$$

and

$$\tilde{G}_{\ell j} = \frac{1}{n} \sum_{k=1}^n (\hat{G}_{\ell k} + \hat{G}_{kj})$$

which, on adding together immediately, gives

$$\tilde{G}_{i\ell} + \tilde{G}_{\ell j} = \frac{1}{n} \sum_{k=1}^n (\hat{G}_{ik} + \hat{G}_{kj})$$

where use has been made of the fact that $\hat{G}_{k\ell} = -\hat{G}_{\ell k}$. This equation immediately gives

$$\tilde{G}_{ij} = \tilde{G}_{i\ell} + \tilde{G}_{\ell j}$$

which demonstrates that $\tilde{G}_{ij}$ defined in (18) will always satisfy the transitivity axiom 4.

Similarly, it can be easily proved that $\tilde{I}_{ij}$ will always satisfy axiom 5. Thus, we have demonstrated that our proposed measures $\tilde{G}_{ij}$ and $\tilde{I}_{ij}$ will always satisfy all the axioms proposed in the paper.

6. Explaining Changes in Poverty in Thailand

The methodology developed in this paper will now be applied to analyze changes in poverty in Thailand. We make use of the four socio-economic surveys conducted in Thailand during the years 1988, 1990, 1992 and 1994. These are nationwide surveys covering all private, non-institutional households residing permanently in municipal areas, sanitary districts and villages.¹

To analyze poverty, we need to measure the economic welfare of each household in the society. In this study, we measured the household welfare by its per capita income. The income concept used is fairly comprehensive consisting of both money and inkind income.

Since households differ in size, age, composition and many other characteristics, it is expected that they will have differing needs. Clearly then, a single poverty line cannot be used for all households. Kakwani and Krongkaew (1997) have constructed new poverty lines for Thailand which take account of the differing needs of people living in a household. In the process of formulating these poverty lines, they took into account nutritional requirements of Thai population, food consumption patterns of Thai population and price differences between

¹ These are the administrative areas of the Thai public administration system. Municipal areas are the most urbanised in terms of living conditions whereas the villages are the most rural with sanitary districts somewhere in between.

the regions and areas. In this paper, we measured the household welfare by the ratio of per capita income of a household to the per capita poverty line income of that household. This measure of welfare can be interpreted as the percentage of excess income households have over their poverty lines and takes into account the differing needs of households and the price differences among the regions and areas of Thailand.

Let x_i be the per capita income of the i th household and z_i the poverty threshold for that household, then the household welfare is given by $y_i = x_i/z_i$. The i th household is classified as poor if $y_i < 1$ and non-poor otherwise. Let us define a variable

$$I_i = 1, \text{ if } y_i < 1$$

$$= 0, \text{ if } y_i \geq 1$$

then the head-count measure of poverty H is given by

$$H = \sum_{i=1}^n a_i I_i$$

where a_i is the proportion of persons living in the i th household (a_i is obtained from the population weight attached to the i th sample household), is the proportion of poor persons in Thailand. An alternative measure of poverty is the poverty gap ratio which is defined as

$$g = \sum_{i=1}^n a_i I_i (1 - y_i)$$

This measure will provide adequate information about the intensity of poverty if all the poor are assumed to have the same y_i . If y_i is not the same for all the poor (which is always the case), then one can have a class of measures

$$FGT = \sum_{i=1}^n a_i I_i (1 - y_i)^\alpha$$

where $\alpha > 1$. This is the class of poverty measures suggested by Foster, Greer and Thorbecke (1984).

The FGT measures of poverty distinguish among the poor according to how far below the poverty line their income falls and can, therefore, be considered as measures of depth of poverty. In this paper, it will suffice to use the FGT measure for which $\alpha = 2$.²

Table 1 presents the estimates of average welfare and poverty in Thailand for the years 1988, 1990, 1992 and 1994. The main results of our proposed decomposition are presented in Table 2.

It can be noted from Table 1 that the average welfare in Thailand has increased substantially between 1988 and 1994; it increased from 211.90 percent points in 1988 to 343.86 percent points in 1994. These increases are quite spectacular. The benefits of economic growth are indeed flowing to household and then to individuals. This is also indicated by the reduction in poverty as measured by the head-count and poverty gap ratios. The percentage of poor decreased from 32.59 in 1988 to only 16.32 in 1994. A similar pattern emerges when we measure poverty by the poverty gap ratio which decreased from 10.39 percent points in 1988 to 4.28 percent points in 1994. Unfortunately, the FGT index does not show a monotonic reduction in poverty. As a matter of fact, this index shows an increase in poverty in the periods 1988-90, 1990-92. In the last period 1992-94, the index reduced somewhat but not by much.

The growth and inequality effects were computed by two alternative procedures. One procedure is based on bilateral comparisons and another on multilateral comparisons. The estimates by the two methods are different but not by much. The procedure based on multilateral comparisons is of course more desirable because it satisfies all the intuitive natural axioms (the most important among them being the transitivity axioms).

² There exists a large literature on poverty measures (see Sen 1976, Kakwani 1980). For a survey of various measures, see Kakwani (1997). We believe that the three measures used here provide sufficient information about the important attributes of poverty.

It can be observed that the inequality effect is positive for all the periods except for the last one (1992-94) when it is negative for the head-count and poverty gap ratios. These results imply that the redistribution of income which has occurred with the economic growth has benefited the rich more than the poor. As a result, the economic growth has contributed to much less reduction in poverty. For instance, if inequality did not change, the percentage of poor would have reduced by 20.65 percent points in the 1988-94 period but the actual reduction is 16.27 percent points. Thus, the change in inequality has contributed to an increase in the percentage of poor by 4.39 percent points in the same period. The reduction in poverty measured by the head-count ratio is maximum in the 1992-94 period which is mainly due to the fact that the redistribution of income has contributed to a reduction in poverty (instead of an increase as is the case for other years).

For both the head-count and poverty-gap ratios, the growth effect has dominated over the inequality effect which has resulted in a substantial reduction in poverty. But this is not the case when we measure poverty by the Fost-Greer-Thorbecke index. For this index, the adverse effect of the inequality effect has dominated the favourable growth effect, as a result, poverty has increased monotonically between 1988 and 1992.

The FGT index measures the depth of poverty and is highly sensitive to the income among the very poor. The increasing value of this index demonstrates that the ultra-poor in Thailand may have become worse off in spite of a high economic growth. This is an important finding. It suggests that the government of Thailand cannot be complacent with an impressive growth performance. The government has to play a more active role in targeting the ultrapoor. The economic growth alone does not benefit the ultrapoor.

Table 1:

Period	Average welfare	Head-count ratio	Poverty gap ratio	FGT* index
1988	211.90	32.5893	10.39	1.1119
1990	254.01	27.217	7.9683	1.3361
1992	295.06	23.1669	6.7566	1.8095
1994	343.86	16.32	4.2854	1.6832

* Foster

Table 2:

Years	Total change in poverty	Bilateral		Multilateral	
		Growth component	Inequality component	Growth component	Inequality component
Head count					
1988-90	-5.37	-8.68	3.31	-8.34	2.97
1988-92	-9.42	-15.06	5.64	-15.06	5.64
1988-94	-16.27	-20.31	4.04	-20.65	4.39
1990-92	-4.05	-7.01	2.96	-6.72	2.67
1990-94	-10.90	-12.37	1.48	-12.32	1.42
1992-94	-6.85	-5.89	-0.96	-5.60	-1.25
Poverty gap					
1988-90	-2.42	-3.73	1.31	-3.63	1.21
1988-92	-3.63	-6.31	2.67	-6.20	2.57
1988-94	-6.10	-7.96	1.86	-8.17	2.07
1990-92	-1.21	-2.65	1.44	-2.57	1.36
1990-94	-3.68	-4.56	0.88	-4.54	0.86
1992-94	-2.47	-2.16	-0.31	-1.97	-0.50
Foster-Greer -					
1988-90	0.22	-1.89	2.12	-1.88	2.10
1988-92	0.70	-3.21	3.91	-3.14	3.84
1988-94	0.57	-3.94	4.51	-4.03	4.60
1990-92	0.47	-1.28	1.75	-1.26	1.73
1990-94	0.35	-2.14	2.49	-2.15	2.50
1992-94	-0.13	-0.99	0.86	-0.89	0.77

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